## RLC Series Circuit and Resonance

**1. RLC Series Circuit**

An RLC series circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series with an AC voltage source. The same current flows through all three components, but the voltage across each component may differ in magnitude and phase.

**2. Function of Each Component**

1. Resistor (R): Opposes the flow of current and dissipates electrical energy as heat.
2. Inductor (L): Opposes changes in current and stores energy in its magnetic field.
3. Capacitor (C): Stores energy in its electric field and opposes changes in voltage.

**3. Reactance**

Reactance is the opposition offered by an inductor or capacitor to alternating current (AC).

Inductive Reactance:

$X_L = 2\pi fL$

Capacitive Reactance:

$X_C = 1/(2\pi fC)$

Where:

- $X_L$ = Inductive reactance (Ω)
- $X_C$ = Capacitive reactance (Ω)
- f = Frequency (Hz)
- L = Inductance (H)
- C = Capacitance (F)

**4. Impedance of an RLC Series Circuit**

Impedance is the total opposition offered by a series RLC circuit to alternating current.

Formula:

$Z = \sqrt{[R^2 + (X_L - X_C)^2]}$

Where:

- Z = Impedance (Ω)
- R = Resistance (Ω)
- $X_L$ = Inductive reactance (Ω)
- $X_C$ = Capacitive reactance (Ω)

The current in the circuit is given by:

$I = V/Z$

Where V is the applied RMS voltage and I is the RMS current.

**5. Resonance in a Series RLC Circuit**

Resonance is the condition in which the inductive reactance becomes equal to the capacitive reactance. At this condition, XL = XC, and their effects cancel each other.

The circuit behaves like a purely resistive circuit.

Condition of Resonance:

XL = XC

$2\pi frL = 1/(2\pi frC)$

Resonant Frequency:

$fr = 1/(2\pi\sqrt{LC})$

Where fr is the resonant frequency in hertz (Hz).

**6. Characteristics of Series Resonance**

1. Inductive reactance is equal to capacitive reactance (XL = XC).
2. Impedance becomes minimum and is equal to resistance (Z = R).
3. Current becomes maximum (I = V/R).
4. The phase angle between voltage and current is zero.
5. The power factor becomes unity ($\cos \phi = 1$).
6. The circuit behaves as a purely resistive circuit.
7. The voltages across the inductor and capacitor are equal in magnitude and opposite in phase.

**7. Behaviour of the Circuit at Different Frequencies**

| Condition | Reactance | Circuit Behaviour |
|---|---|---|
| f < fr | XC > XL | Capacitive |
| f = fr | XC = XL | Resistive |
| f > fr | XL > XC | Inductive |

**8. Quality Factor (Q-Factor)**

The quality factor indicates the sharpness of resonance in an RLC series circuit. A higher Q-factor means sharper resonance and a narrower bandwidth.

Formula:

$Q = XL/R = (2\pi frL)/R$

Alternatively,

$Q = 1/(2\pi frCR)$

**9. Bandwidth**

Bandwidth is the difference between the upper and lower half-power frequencies of a resonant circuit.

Formula:

$BW = f_2 - f_1$

For a series RLC circuit:

$BW = R/(2\pi L)$

Also,

$Q = fr/BW$

Where:

- $f_1$ = Lower half-power frequency
- $f_2$ = Upper half-power frequency
- fr = Resonant frequency
- BW = Bandwidth in hertz (Hz)

**10. Applications of Series Resonance**

1. Used in radio and television tuning circuits.
2. Used in frequency-selective circuits.
3. Used in communication systems to select a desired frequency.
4. Used in filters and electronic measuring instruments.

**Conclusion**

A series RLC circuit contains resistance, inductance, and capacitance connected in series. At resonance, inductive reactance equals capacitive reactance, resulting in minimum impedance, maximum current, and unity power factor. Resonance is useful for selecting a particular frequency in electrical and electronic circuits.